

MINOTHMA SITHUMINI

Software Engineering Intern

+94 71 931 9477 | minothmasithumini@gmail.com | linkedin.com/in/minothma | github.com/Minothma | Ratnapura, Sri Lanka

Professional Summary

An IT undergraduate at the University of Moratuwa with hands-on experience in full-stack development, building end-to-end web applications using Next.js, React, Spring Boot, and NestJS. Skilled in designing RESTful APIs, relational database systems, and secure authentication workflows through academic and personal projects. Actively engaged in university leadership and volunteer initiatives, currently seeking a Software Engineering Internship to apply technical skills and contribute to real-world projects.

Education

B. Sc. (Hons) in Information Technology and Management (CGPA: 3.47 / 4.00) 2024 - Present

University of Moratuwa, Sri Lanka

G.C.E Advanced Level 2022

Sumana Balika Vidyalaya, Ratnapura

- Physical Science Stream – Combined Mathematics (A), Chemistry (B), Physics (C) – Z-score: 1.2516

Technical Skills

Programming Languages: Java, C, Python, JavaScript, TypeScript, PHP, SQL

Frontend Development: HTML, CSS, React, Next.js, Tailwind CSS

Backend Development: Spring Boot, RESTful APIs

Databases: PostgreSQL, MySQL

DevOps & Tools: Docker, Git, Postman

UI/UX & Design: Figma, GIMP

Concepts: OOP, Data Structures & Algorithms, Authentication & Authorization, Database Design, RESTful API Design, Agile Methodology

Projects

Durdans Hospital LIMS | Full-Stack Developer, 2nd-Year Group Project Aug 2025 – Aug 2026

- Tech:** Next.js, React, TypeScript, Spring Boot, PostgreSQL, Keycloak
- Built end-to-end laboratory modules – Sample Reception, Accessioning, MLT Result Entry, and Quality Control – using Next.js and Spring Boot.
- Developed an automated sample verification system with tube-cap recognition, rejection workflows, barcode reprints, and priority triage (STAT / Urgent / Routine).
- Implemented real-time result validation with dynamic reference range checks, longitudinal delta-checks, and Westgard QC rules.
- Integrated Keycloak RBAC and PostgreSQL audit logs to secure RESTful APIs and ensure full specimen traceability.
- Designed responsive, real-time lab dashboards with live Turnaround Time (TAT) tracking and instant alerts for critical/panic test results.

Job Application Tracker | Full-Stack Developer, Individual Project Aug 2026 – Sep 2026

- Tech:** Next.js, React, TypeScript, NestJS, PostgreSQL, Prisma, Flyway, AWS S3, Docker
- Developed an end-to-end application lifecycle tracking platform featuring an interactive drag-and-drop Kanban board to manage stages from Applied to Offer and Rejection.
- Built multi-version resume management with direct AWS S3 presigned uploads, interview scheduling, and activity notes timeline, streamlining the job application workflow.
- Architected modular RESTful APIs in NestJS with JWT authentication (Access & Refresh tokens), integrated Flyway database migrations with Prisma ORM, and containerized the setup using Docker.
- Designed a responsive dashboard with real-time application status analytics and progress tracking across multiple job applications.

- **Tech:** PHP, MySQL, JavaScript, HTML, CSS, Bootstrap, AJAX
- Built a full-stack blogging platform with user authentication and full CRUD functionality, featuring rich-text publishing, draft auto-saving, and SEO-friendly URLs.
- Developed interactive features including real-time AJAX search, a multi-type reaction system, and a dynamic comment section to boost user engagement.
- Strengthened platform security with CSRF protection, SQL injection prevention (prepared statements), and created an Admin Dashboard for user and content moderation.

Battery Vitals | Embedded Systems Developer, 1st-Year Group Project

Aug 2024 – Aug 2025

- **Tech:** Python, ESP32, Embedded Systems
- Implemented automated charging, resting, and discharging cycles for Li-Po batteries using ESP32 and relay-based switching.
- Integrated TP4056, TP5100, and IP2312 charging modules with configurable discharge loads to generate battery cycle data.
- Implemented temperature monitoring and fan-based thermal protection using DS18B20, and contributed to SOH analysis through charge-discharge curve evaluation.

Certifications

• Cloud Essentials – IBM	May 2025
• Python for Data Science – IBM	Mar 2025
• SQL and Relational Databases 101 – IBM	Mar 2025
• SQL (Advanced) – HackerRank	Apr 2025
• Web Design for Beginners – University of Moratuwa	Feb 2025
• Python for Beginners – University of Moratuwa	Feb 2025
• Peritus Research Bootcamp – AI/ML & Data Science – Peritus Research	In Progress, Expected Feb 2027

Leadership & Activities

- **Director of IT** – Rotaract Club of Alumni of University of Moratuwa (Term 26/27)
- **Editorial Committee Member (Term 25/26)** – IEEE RAS Student Branch Chapter, University of Moratuwa
- **Committee Member (Term 25/26)** – IEEE Communications Society Student Branch Chapter, University of Moratuwa
- **Committee Member** – Cancer Awareness Campaign 2025 & Hand in Hand Sub-Project, Rotaract Club of University of Moratuwa
- **Finance Committee Member** – HackElite 2.0, IEEE WIE Student Branch Affinity Group, University of Moratuwa
- **Editorial Committee Member** – Rise Up Mora 2025, IEEE Student Branch, University of Moratuwa

Languages

English, Sinhala (Native)

References

Mr. B.H. Sudantha

Former Dean & Senior Lecturer
Faculty of IT, University of Moratuwa
Email: sudanthabh@uom.lk
Phone: +94 76 927 1676

Mr. P.A. Dasith Niwanthaka

Software Engineer
Codegen International Pvt Ltd
Email: niwanthaka77@gmail.com
Phone: +94 70 396 9096